

5058 Project 1: Time Series Analysis with Financial Data

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Abstract. This project presents a comprehensive time series analysis of the daily log-returns for Apple Inc. (AAPL) and Microsoft Corp. (MSFT) from 2008 to 2023. The study transitions from classical linear frameworks to advanced non-linear and multi-scale methodologies to characterize market dynamics. Initial diagnostics using Augmented Dickey-Fuller (ADF) tests confirm the stationarity of the return series, while a VARMA(1,1) model reveals a lack of significant lagged cross-correlations, suggesting weak-form market efficiency despite strong contemporaneous correlation. In the frequency domain, Power Spectral Density (PSD) analysis characterizes the returns as predominantly white noise. However, by employing Empirical Mode Decomposition (EMD), the research identifies hidden structural components across different time scales. The analysis further utilizes Rescaled Range (R/S) analysis and Detrended Fluctuation Analysis (DFA) to estimate Hurst exponents, finding that while the raw series resemble random walks ($H \approx 0.5$), specific intrinsic mode functions (IMFs) exhibit strong persistence ($H > 0.9$). Finally, a Multifractal DFA (MFDFA) approach confirms the presence of multifractality in both assets, demonstrating that market volatility is driven by a complex spectrum of scaling exponents rather than a single fractal dimension. These findings underscore the limitations of linear models and highlight the necessity of multi-scale, non-linear techniques in financial risk modeling.

Keywords: Financial Time Series · VARMA Model · Empirical Mode Decomposition (EMD) · Hurst Exponent · Detrended Fluctuation Analysis (DFA) · Multifractality · Market Efficiency.

1 Data Preprocessing

This section uses AAPLe Inc. (AAPL) and Microsoft Inc. (MSFT) as research subjects. Historical daily closing price data was downloaded from Yahoo Finance's 'yfinance' interface. To meet the requirement of more than 4000 trading days, the research period was set to all trading days from 2008 to 2023.

The aligned daily closing price sequences are denoted as $S_1(t)$ and $S_2(t)$, respectively. The daily logarithmic return is defined as:

$$X_i(t) = \ln \left(\frac{S_i(t)}{S_i(t-1)} \right), \quad i = 1, 2. \quad (1)$$

After calculating the returns, the sequences are reduced in dimensionality by subtracting the sample mean to ensure that the mean of each updated return sequence is zero.

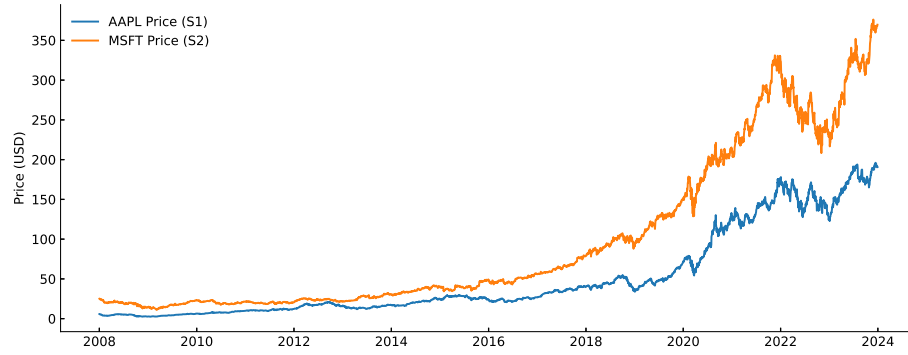


Fig. 1. Stock Price of AAPL & MSFT

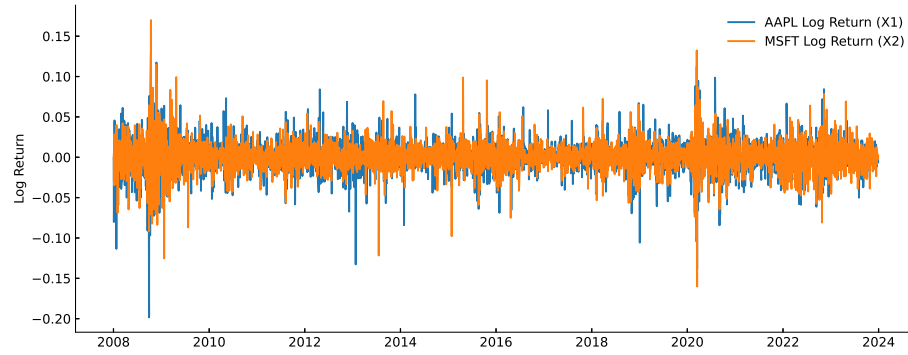


Fig. 2. Log Return of AAPL & MSFT

2 Stationarity and Autocorrelation

2.1 Stationarity Testing

Perform an ADF Test on X to judge its stationarity.

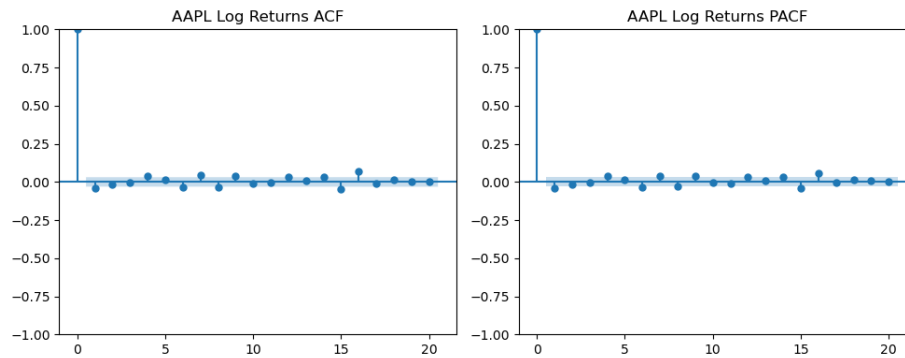
Table 1. ADF Test Result

	AAPL	MSFT
ADF Statistic	-15.12	-22.81
# Lags	15	8
# Obs	4010	4017
Critical 1%	-3.43	-3.43
Critical 5%	-2.86	-2.86
Critical 10%	-2.57	-2.57
Stationary	Yes	Yes

The results of the ADF test show that both log returns are stationary sequences.

2.2 Linear Autocorrelation and ARMA Identification

Do ACF and PACF Analysis, then select the best ARMA model. See Fig. 3 and Fig. 4

**Fig. 3.** ACF & PACF of AAPL

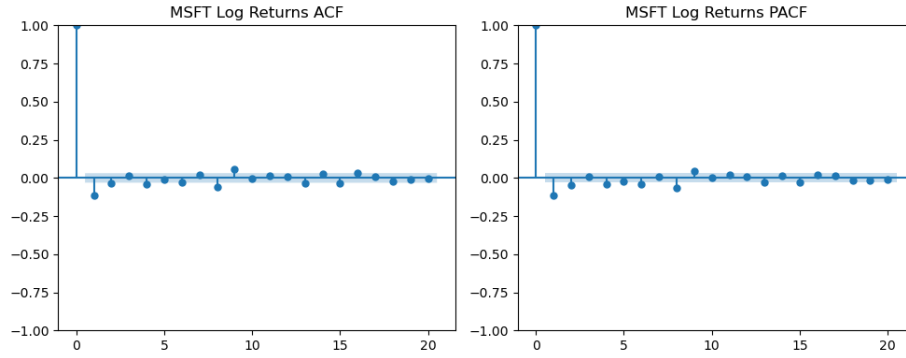


Fig. 4. ACF & PACF of MSFT

For both stocks, since most orders of ACF/PACF are not significant, the return series conforms to the weak-form efficient market hypothesis. It is recommended to choose ARMA(0,0) or a very low-order model.

Combined with the previous ADF test, this indicates that the stock price fluctuations are relatively smooth, with no significant long-term upward or downward trend; the daily price movements are almost random.

2.3 Non-linear Autocorrelation Analysis

Plot the ACF of $|X|$ (absolute returns) and compare the trends of the two autocorrelations. See Fig. 5

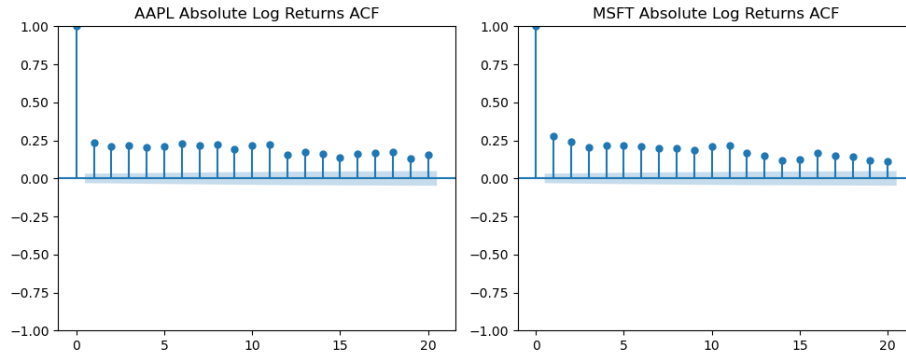


Fig. 5. ACF of absolute log returns

This phenomenon indicates that the return series itself has almost no linear correlation (i.e., the direction is unpredictable, close to an efficient market), but

its volatility exhibits a significant nonlinear correlation, reflecting the typical **volatility clustering** characteristic in financial markets.

3 Fractal Behavior of Time Series

3.1 Hurst Exponent

This section calculates the Hurst exponent of the sequence using R/S analysis and plots a scatter plot of $\log(n)$ versus $\log(R/S)$.

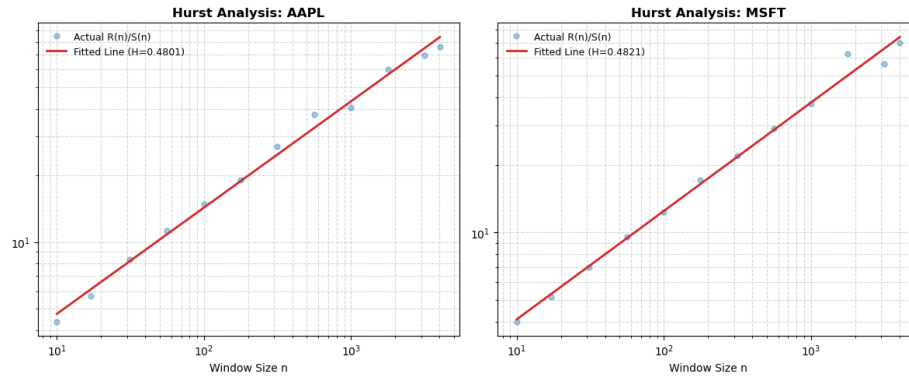


Fig. 6. Hurst Analysis

The log returns of AAPL and MSFT are predominantly stochastic and memoryless, behaving as nearly perfect Brownian motion with no significant long-term trending or mean-reverting bias.

3.2 Detrended Fluctuation Analysis (DFA)

DFA is an alternative method to Hurst Exponent for detecting long-range correlations in non-stationary time series. It is often considered more robust because it removes local trends that can lead to a false detection of long-range memory.

The scaling exponent α is theoretically equivalent to the Hurst exponent (H) for stationary noise-like series (such as log returns).

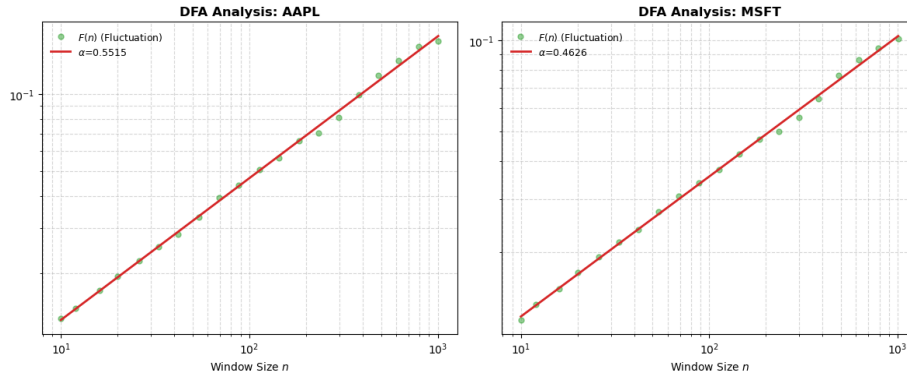


Fig. 7. DFA Analysis

The scaling exponent α calculated via DFA is 0.5515 for AAPL and 0.4526 for MSFT. These values represent the fluctuation scaling behavior of the log-return series across different time scales. These values are consistent with the previously calculated Hurst exponents ($H = 0.4801$ and $H = 0.4821$). For both stocks, α and H remain close to the theoretical value of 0.5, which characterizes a Random Walk (Brownian Motion). This confirms that the return series for both assets are largely memoryless and behave similarly to a stochastic process with very weak long-range correlation.

3.3 Multifractality

Previous methods like the Hurst index and DFA actually assumed that the sequence was monofractal, meaning that self-similarity was constant across all scales and volatility levels. However, financial data (such as stock prices) typically exhibits multifractal characteristics, meaning that the scaling patterns of large and small fluctuations differ.

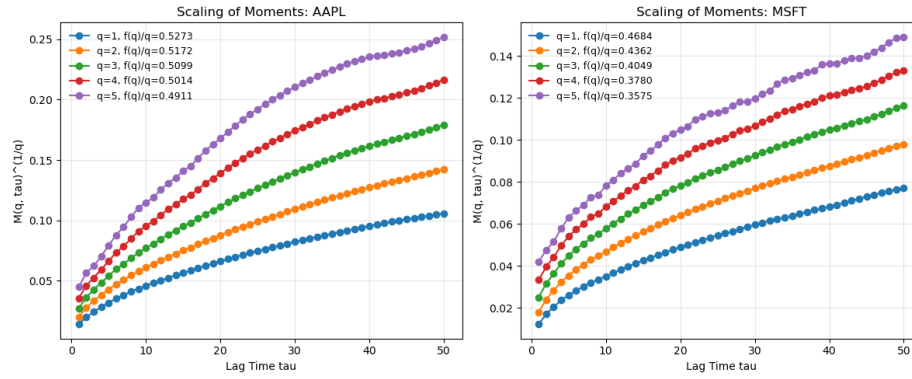
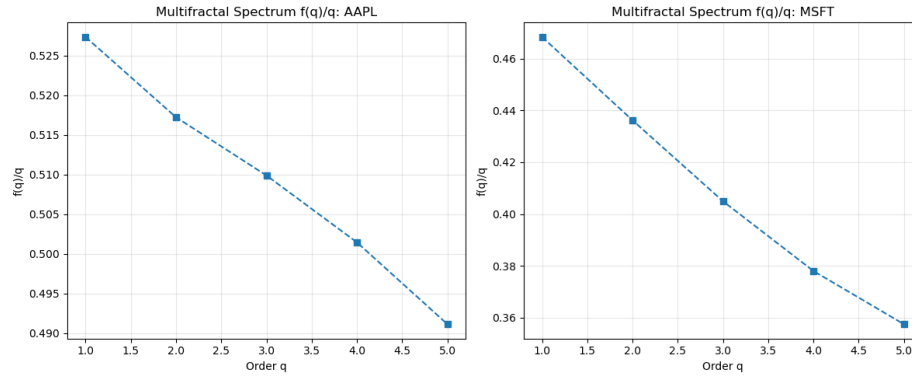


Fig. 8. Scaling of Moments

Fig. 9. $f(q)/q$ Spectrum

Asset S (and return X) are multifractal, and experimental results show that both $\alpha(q)$ and $f(q)/q$ decrease as q increases (AAPL and MSFT). This indicates that large fluctuations ($q > 0$) and small fluctuations ($q < 0$) follow different scaling logics, which is a typical characteristic of multifractal systems.

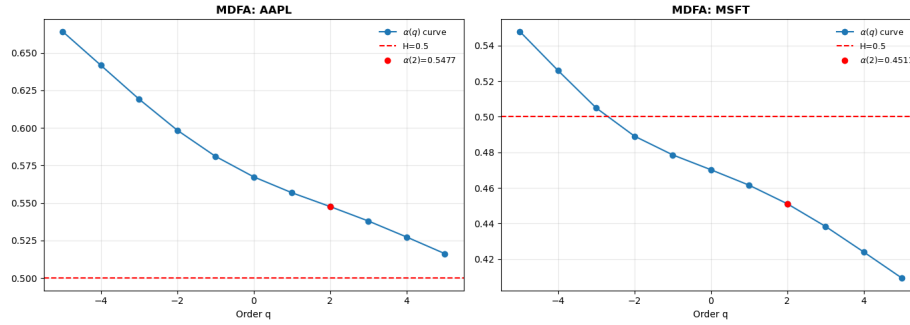


Fig. 10. MDFA Analysis

The results from MDFA are consistent with those from the moment-based method (Eq. 2). In terms of trends, the spectra obtained by both methods show a downward trend, both pointing to the multifractal conclusion. Numerically, for MSFT, $f(q)/q$ is 0.4362 at $q = 2$, while $\alpha(2)$ for MDFA is 0.4511; for AAPL, $f(q)/q$ is 0.5172 at $q = 2$, and $\alpha(2)$ is 0.5477. Considering the noise in the numerical calculations, the magnitudes and trends of the two methods are completely consistent.

4 Granger Causality

To capture the potential linear interdependencies between the log-returns of AAPL and MSFT, we estimated a **VARMA(1,1)** model. The model diagnostics and estimated coefficients are summarized in Table 2.

Table 2. VARMA(1,1) Model Estimation Results

Metric / Variable	Coef	Std Err	z	P > z
<i>Model Diagnostics</i>				
Log Likelihood	22249.29	AIC	-44472.58	
Prob(Q) [AAPL, MSFT]	1.00, 0.99	BIC	-44390.21	
<i>Equation: AAPL</i>				
Intercept	6.06e-06	0.000	0.019	0.985
L1.AAPL	0.0128	0.388	0.033	0.974
L1.MSFT	-0.0645	0.185	-0.349	0.727
<i>Equation: MSFT</i>				
Intercept	4.19e-07	0.000	0.001	0.999
L1.AAPL	-0.0628	0.351	-0.179	0.858
L1.MSFT	-0.0550	0.176	-0.313	0.754
<i>Error Covariance</i>				
sqrt.var.AAPL	0.0196	0.000	175.60	0.000
sqrt.cov.AAPL.MSFT	0.0104	0.000	74.225	0.000
sqrt.var.MSFT	0.0144	6.86e-05	210.25	0.000

Choose (1,1) as parameter of VARMA model. The Ljung-Box (Q) test results showed P-values (Prob(Q)) of 1.00 and 0.99, respectively. This means that the model has successfully extracted all linear correlations in the residuals, and the residual sequence is close to white noise.

At a significance level of 5%, all lagged regression coefficients in this model (both individual and cross-correlation coefficients) are insignificant.

Since the lag coefficient L1.MSFT in the AAPL equation has a P value of 0.727, which is much greater than 0.05, cannot reject the null hypothesis that "MSFT is not a Granger cause of AAPL". Similarly, the lag coefficient L1.AAPL in the MSFT equation has a P value of 0.858, which also fails to reject the null hypothesis. Based solely on the results of the VARMA(1,1) model, there is no significant Granger causality between AAPL and MSFT.

The p-value of $\text{sqrt.cov.AAPL.MSFT}$ is 0.000, which is highly significant. This indicates a strong concurrent correlation between AAPL and MSFT. In other words, they are linked synchronously, rather than being driven by a sequential causal relationship. This implies that they are both driven in real-time by macroeconomic factors such as Nasdaq index fluctuations or Federal Reserve policies.

5 Fourier Transform and Power Spectrum

To analyze the frequency-domain characteristics of the stock return series, I first computed the Discrete Fourier Transform (DFT) of the log-returns (X) and plotted the magnitude of the resulting Fourier coefficients against frequency. Following this, the Power Spectral Density (PSD) was estimated and visualized. In accordance with the Nyquist-Shannon sampling theorem, the frequency range was constrained to a maximum of 0.5 cycles per day for the daily time series.

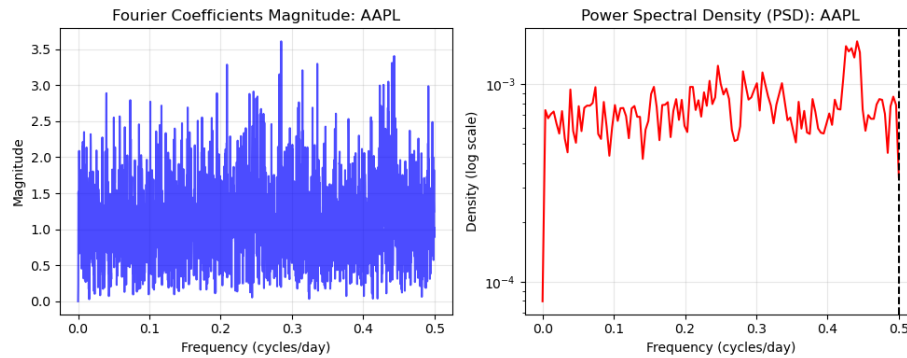


Fig. 11. FT and PSD of AAPL

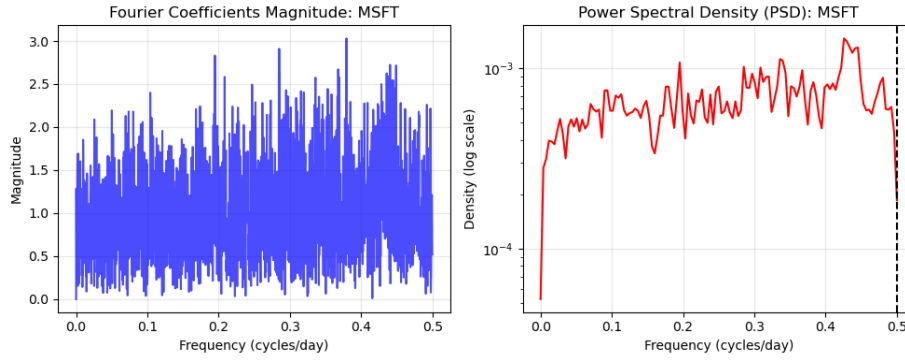


Fig. 12. FT and PSD of MSFT

From White Noise Characteristics: The power spectral density (PSD) curves for both AAPL and MSFT are largely flat across the entire frequency range. This indicates that the return series are approximately White Noise, with power distributed almost equally across all frequencies. This is consistent with the efficient market hypothesis (EMH), suggesting returns are not predictable based on past frequencies.

From Periodicity: there are no sharp, dominant peaks in either the Magnitude or PSD plots. Therefore, we can comment that the returns do not exhibit any strong, fixed periodicities (like a strong weekly or monthly cycle).

The flat PSD confirms the previous findings of a Hurst exponent close to 0.5, representing a memoryless random process.

6 Empirical Mode Decomposition

AAPL decomposed into 10 IMFs, MSFT decomposed into 11 IMFs.

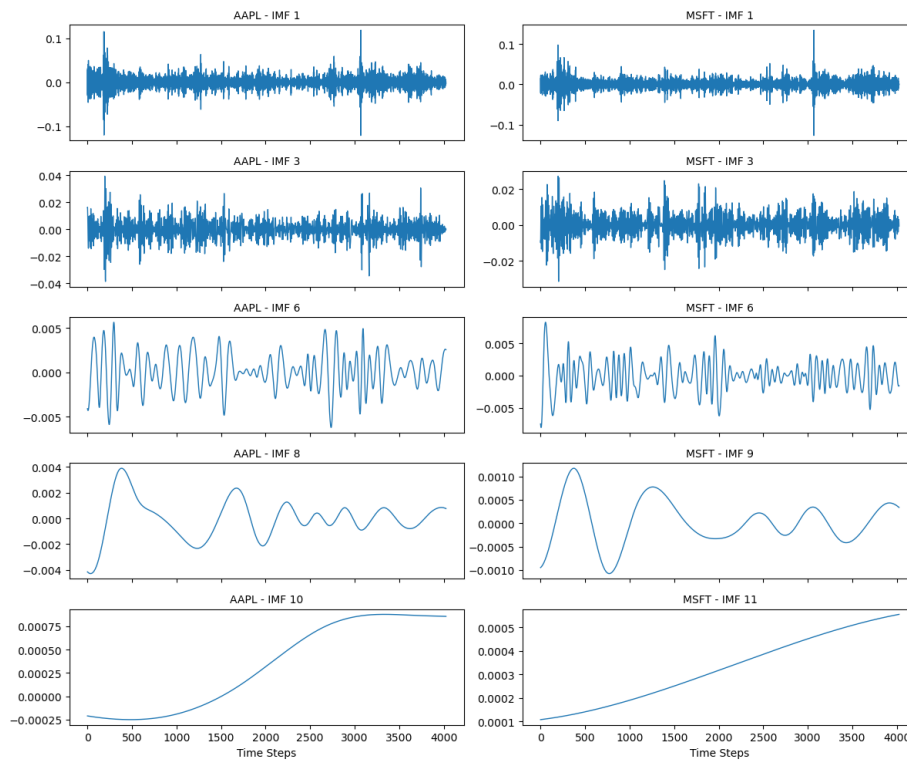


Fig. 13. IMF Decomposition

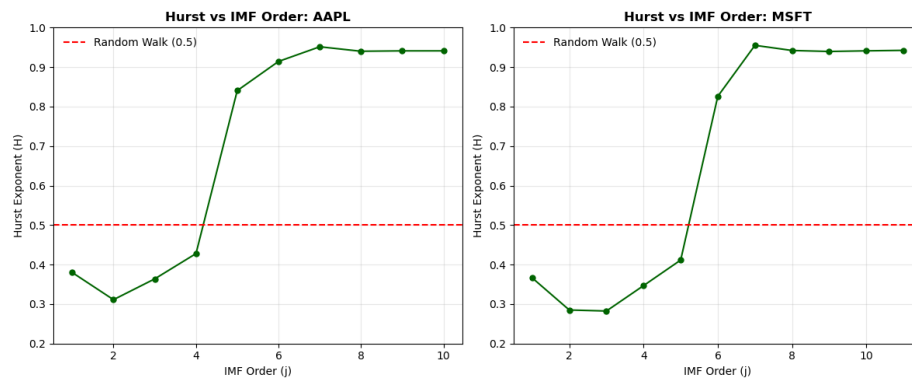


Fig. 14. Hurst vs IMF order

The trend shows a clear transition from anti-persistent ($H < 0.5$) behavior in lower-order IMFs to highly persistent ($H > 0.9$) behavior in higher-order IMFs.

- Lower Orders ($j=1$ to 4): These components represent high-frequency market noise and volatility, showing a mean-reverting nature.
- Transition: There is a sharp increase in the Hurst exponent around $j=5$, representing the filter shifting from noise to structural signals.
- Higher Orders ($j > 6$): The Hurst exponent stabilizes near 0.95 , indicating that the long-term underlying trends of both AAPL and MSFT are extremely smooth and possess strong positive auto-correlation (long memory).

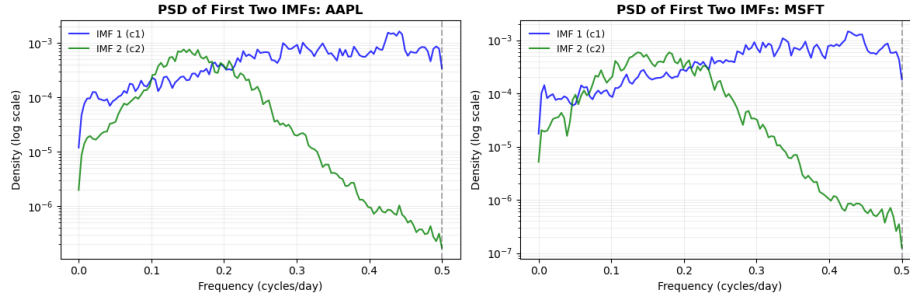


Fig. 15. PSD vs 2 IMFs

The PSD plots of the first two IMFs exhibit the characteristics of adaptive band-pass filters.

- IMF 1 (c_1) acts as a high-frequency extractor, with its power concentrated in the $0.25 - 0.5$ cycles/day range, capturing the most rapid oscillations (noise) in the stock returns.
- IMF 2 (c_2) shows a peak shifted to the left (lower frequency), primarily covering the $0.1 - 0.2$ cycles/day band.

Visually, they look like a dyadic filter bank where each successive IMF captures a lower frequency scale than the previous one. This confirms that EMD effectively decomposes the 'flat' white-noise spectrum of the original returns into distinct, frequency-ordered components.

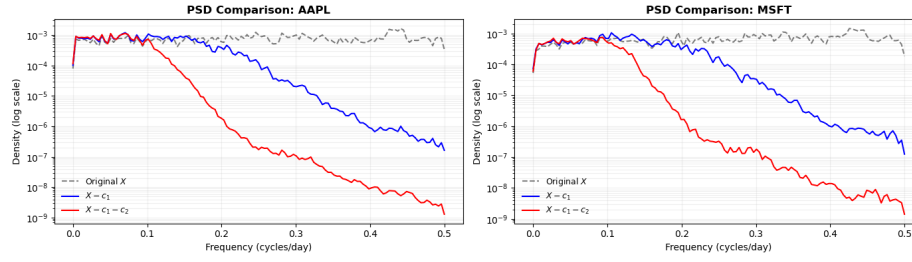


Fig. 16. PSD comparison

The results clearly demonstrate that removing successive IMFs acts as an adaptive low-pass filter. Unlike traditional Fourier filters with fixed cutoff frequencies, EMD identifies the 'noise' floor based on the data's intrinsic scales. As we subtract c_1 and c_2 , the high-frequency 'shimmer' of the stock market is removed, leaving behind a smoother, lower-frequency signal.

Notably, in the very low-frequency range (close to 0), the PSDs of all three series overlap. This suggests that c_1 and c_2 contain zero long-term trend information. They are purely stochastic oscillations. Their removal 'purifies' the series, allowing the underlying long-term cycles or trends to become more dominant in the power spectrum.

The sharp slope of the red and blue lines compared to the flat gray line illustrates the boundary between the unpredictable noise (high frequency) and the potentially structural components (low frequency) of the financial time series. This provides a more concise representation than a standard Fourier Transform, as it concentrates the significant energy into fewer, physically meaningful components.